THE UNIVERSAL FIRST ORDER THICKENING COMMUTES WITH FIBRE PRODUCTS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3250-3267 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 3269-3278, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Universal first order thickenings and their universal property, fibre products of algebraic spaces, and formally unramified morphisms; the verification is retained.

Source reference. Original lemma lemma-base-change-formally-unramified: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2801–2803. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-conormal-functorial-flat: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 557–570. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-universal-thickening-functorial: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3218–3241. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let*

$$\begin{array}{ccc} Z & \xrightarrow{h} & X \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} \\ W & \xrightarrow{h'} & Y \end{array}$$

be a fibre product diagram of algebraic spaces over S with h' formally unramified. Then h is formally unramified and if $W \subset W'$ is the universal first order thickening of W over Y, then $Z = X \times_Y W \subset X \times_Y W'$ is the universal first order thickening of Z over X. In particular the canonical map $f^*\mathcal{C}_{W/Y} \to \mathcal{C}_{Z/X}$ of Lemma lemma-universal-thickening-functorial is surjective.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. The morphism h is formally unramified by Lemma lemma-base-change-formally-unramified. It is clear that $X \times_Y W'$ is a first order thickening. It is straightforward to check that it has the universal property because W' has the universal property (by mapping properties of fibre products). See Lemma lemma-conormal-functorial-flat for why this implies that the map of conormal sheaves is surjective. $\square$